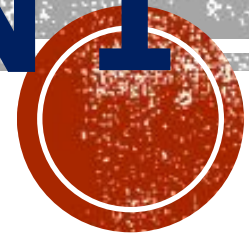




MONTCLAIR STATE
UNIVERSITY

CSIT 416 – IT PROJECT MANAGEMENT



Vaibhav Anu, Instructor
anuv@montclair.edu
Office: CCIS 327D

COURSE PROJECT

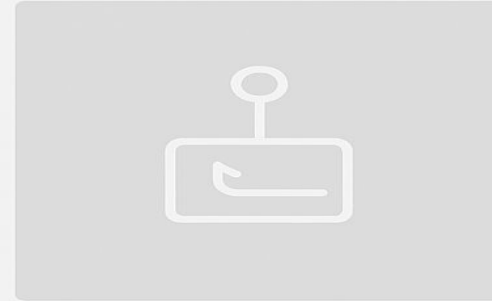
The **PhishGuard** Web Application Development Project

- The web application will be focused on cybersecurity education.
 - It will be designed to:
 - help users learn how to identify phishing attempts,
 - test their skills with quizzes, and
 - defend themselves with practical tools like a phishing scanner.
- The platform will blend education with interactivity, making cybersecurity awareness both engaging and useful.

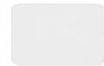
Test your phishing awareness

Learn to identify phishing attacks and protect yourself online

Get Started

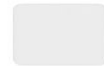


Core Features



Training Modules

Step-by-step lessons on phishing tactics



Quizzes

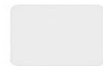
Test your knowledge with interactive quizzes.



Scanner Tool

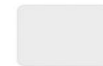
Analyze suspicious emails and URLs

Additional Tools



Knowledge Hub

Articles, tutorials, and resources



Threat Feed

Stay updated on the latest scams

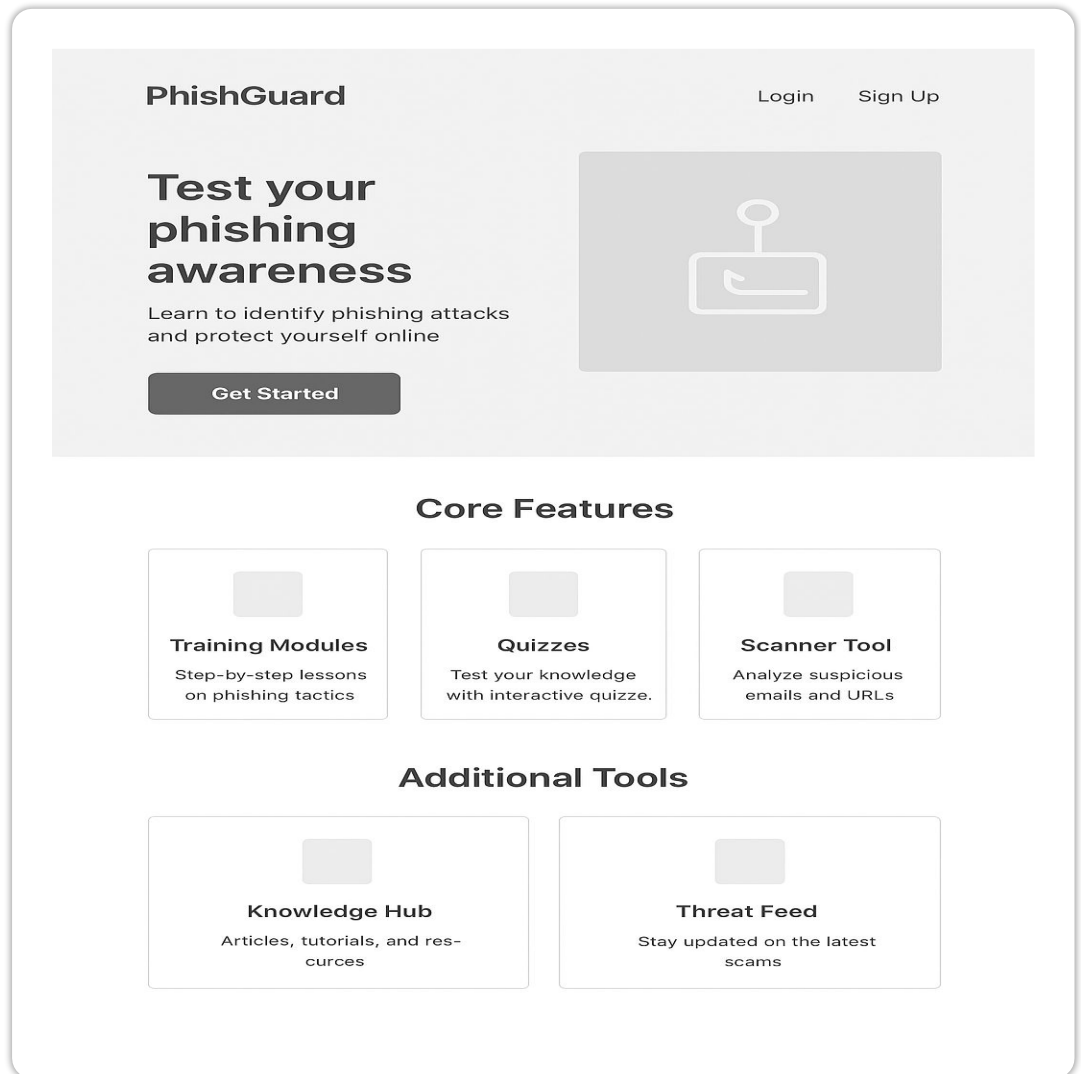
COURSE PROJECT: PHISHGUARD

Project Overview: What is PhishGuard?

- **Cybersecurity awareness web platform**
- Educates users about phishing attacks
- Combines learning, practice, and **real-time tools**
- Engaging, interactive, and practical

HIGH-LEVEL FUNCTIONALITY

- **Training Modules:** Learn phishing tactics step by step
- **Quizzes:** Reinforce skills with interactive challenges
- **Scanner Tool:** Analyze suspicious emails/URLs
- **The AI Component:** Risk score classification for phishing detection

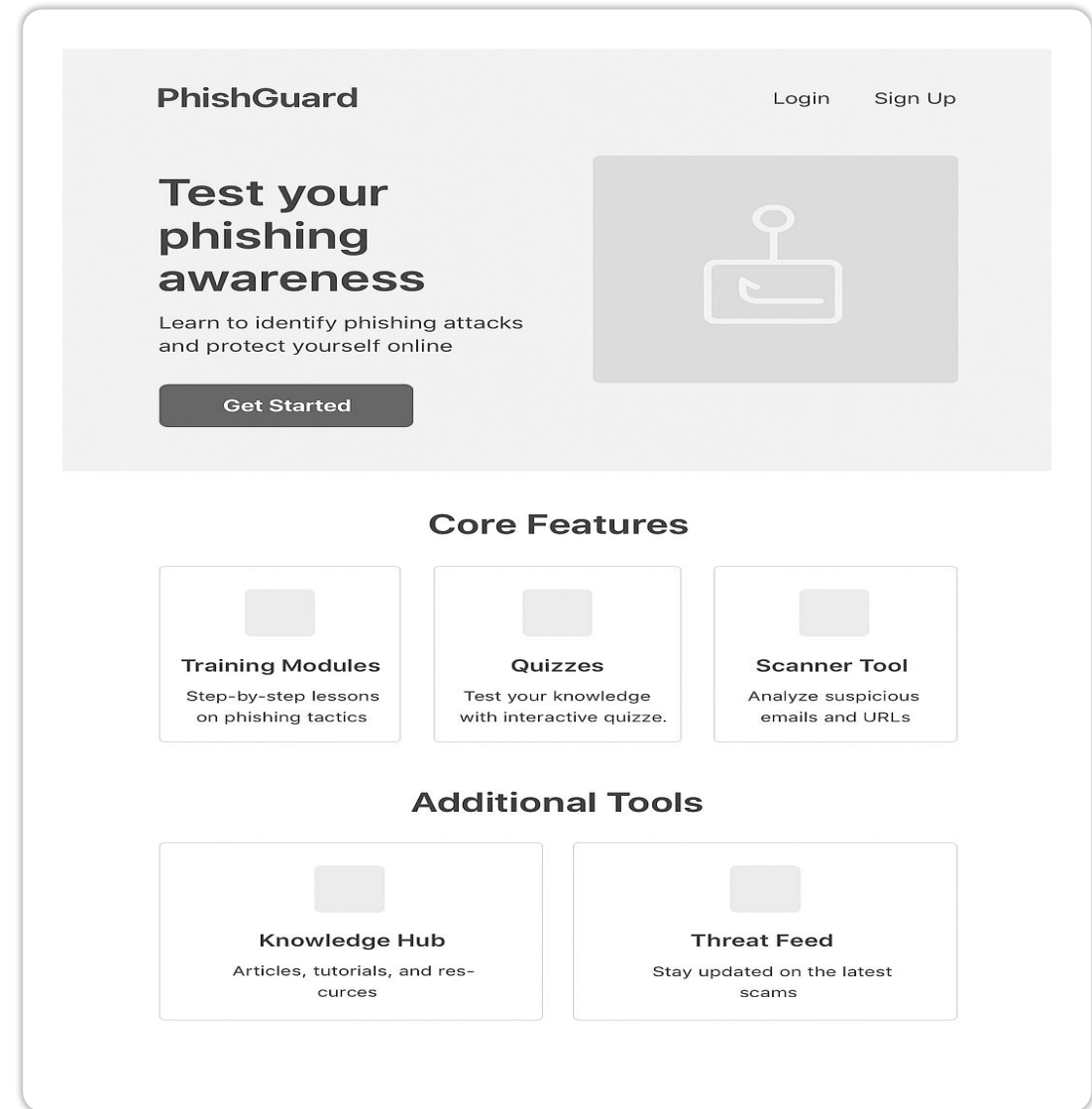


HIGH-LEVEL FUNCTIONALITY

- The heart of PhishGuard revolves around three pillars: Learn, Test, and Defend.
- Users start with training modules, then take quizzes to test their knowledge.
- The phishing scanner lets them apply what they've learned in practice by pasting suspicious content.
- An AI component will classify emails and URLs as phishing or legitimate.

ENGAGEMENT & COMMUNITY

- Leaderboard & badges for progress
- Incident reporting of real phishing attempts
- **Security tools:** password tester, breach checker, checklist
- **Knowledge hub & threat feed** for the latest scams



BASELINE REQUIREMENTS – CORE FEATURES PT.1

1. User Authentication & Profiles

- Register/login
- User dashboard with progress tracking

2. Phishing Awareness Training

- Interactive modules on common attack types (phishing emails, malicious links, password safety)
- Quizzes after each module
- Points awarded for correct answers

3. Suspicious Link/Email Scanner

- Input box where users paste a URL or email text
- System checks against:
 - Blacklisted domains (from database)
 - Suspicious keyword patterns (e.g., “urgent action required”)
- Outputs a risk score (safe/suspicious/high risk)

4. Leaderboard & Gamification

- Leaderboard of top students/users based on quiz scores and challenges
- Badges for completing modules (e.g., “Phishing Pro”, “Social Engineering Buster”)

BASELINE REQUIREMENTS – CORE FEATURES PT.2

5. AI-Powered Email Analysis

- Integrate a simple ML model (e.g., Naive Bayes, or a pretrained transformer) to classify email text as phishing vs. legitimate
- Could start with a small dataset like the Enron email dataset or synthetic phishing datasets

6. Real-Time Alerts / API Integration

- Use an external threat intelligence API (e.g., VirusTotal, PhishTank) to check links/domains against known malicious lists

7. Simulated Phishing Campaigns

- Admins (like instructors) can create fake “phishing emails” inside the system
- Users get tested periodically on whether they can detect them

BASELINE REQUIREMENTS – CORE FEATURES PT.3

8. Knowledge Hub (Resource Library)

- **Content:** Articles, tutorials, infographics, and videos about common attacks (phishing, ransomware, social engineering, password hygiene).
- **Searchable & filterable** by category (email safety, password security, malware, etc.).
- **Bookmark feature** for users to save resources to their profile.

9. Incident Reporting Page

- Users can **report suspicious emails/URLs** they receive.
- Submissions get stored in DB (with optional screenshots/text).
- Other users can upvote/flag as “confirmed phishing.”
- Admins can review and add verified reports to the **blacklist DB**.

SYSTEM REQUIREMENTS – ADDITIONAL FEATURES PT.1

- Security Tools Page
 - A toolkit with:
 - **Password Strength Tester** → checks if a password is strong enough (regex + entropy check).
 - **Breach Checker (API)** → checks if an email appears in data breach lists (e.g., HaveIBeenPwned API).
 - **URL Expander** → reveals shortened links (tinyurl, bit.ly) to show the real domain.